

Speech/Pause Detection Algorithm Based on the Adaptive Method of Complementary Decomposition and Energy Assessment of Intrinsic Mode Functions

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Abstract—Speech/pause detection is one of the important tasks in processing. Its effectiveness depends on the accuracy of measuring amplitude, time, frequency and energy characteristics of speech signals. The main reason for large errors in measurements is due to the use of non-adaptive processing methods. The goal is to develop an algorithm for effective speech/pause detection on the basis of the adaptive method of complementary ensemble empirical mode decomposition (CEEMD). The algorithm is implemented using the adaptive processing method of CEEMD. The adaptability of methods lies in the fact that the basic functions used in the decomposition are extracted from the original speech signal, and allow us to take into account only its inherent features (hidden modulation, energy concentration regions, etc.). To carry out the research of the developed algorithm, a software package for mathematical modeling MATLAB was used. A speech/pause detection algorithm is developed on the basis of the adaptive method of complementary decomposition and energy estimation of empirical modes. A block diagram for the algorithm with a detailed mathematical description is presented. The advantages of the developed algorithm over the known analogs that have gained a wide practical popularity are indicated (STE + ZCR, IE and MFCC). The developed algorithm provides an increase in the correct detection rate of speech/pause by an average of 6%. Comparison of research results with analogs suggests that the developed algorithm is recommended for practical use in voice control systems (VCS).

Keywords—speech signal processing; speech/pause detection; measurement of energy characteristics; complementary ensemble empirical mode decomposition

I. INTRODUCTION

Speech/pause detection is one of the most important tasks in processing of speech commands in VCS. Accurate detection

of the voice command boundaries not only increases the quality of recognition, but reduces the number of computation operations. Today, there are many different approaches for speech/pause detection. The most well-known algorithms are as follows:

- algorithms based on the use of short-time energy values (STE), and the number of short-time signal transitions through zero-crossing rate (ZCR) [1];
- algorithms based on the use of information entropy values (IE) [2];
- algorithms based on the use of mel-frequency cepstrum coefficients (MFCC) [3].

The conducted authors' research [4] of the above-mentioned algorithms has revealed low efficiency in a noisy environment. It is known that the effectiveness of speech/pause detection depends on the correct processing of speech signals, which is determined by the measuring accuracy of its amplitude, time, frequency and energy characteristics. The main reason of large errors in speech/pause detection is associated with the use of non-adaptive processing methods of complex non-stationary and noisy speech signals. In this paper, the authors propose the algorithm for speech/pause detection using the method of adaptive analysis based on the CEEMD [5]. The article is a continuation of the authors' earlier published works [6–9].

II. DESCRIPTION OF THE DEVELOPED SPEECH/PAUSE DETECTION ALGORITHM

Based on of the review of the methods of adaptive analysis, the authors have developed a noise-robust algorithm for VCS, which block diagram is shown in Fig. 1. The essence of the algorithm is the segmentation of the speech signal into

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short fragments for analysis by the CEEMD method with the subsequent formation of a threshold based on the physiological aspect of speech formation, the functional of a human auditory apparatus, and energy assessment of intrinsic mode functions (IMF). Consider the main stages of operation of the proposed algorithm.

Segmentation of the speech signal into fragments is carried out by the following formulas: $S=x(n)/L$, where $x(n)$ is a speech signal, n is discrete timing ($0 < n \leq N$, N is the amount of discrete counting in the signal), S is the amount of fragments in the speech signal, L is the amount of discrete counting in a single fragment; $x_{s+1}(n)=x_i((s \times L)+1; (s+1) \times L)$, where $x_{s+1}(n)$ is the fragment of the speech signal, n is the discrete timing, $s=(0, 1, 2, \dots, S-1)$ is a number of fragment.

An important condition for the noise-robust speech/pause detection is the possibility of an adaptive basis formation, functionally dependent on the content of the source signal. This approach is implemented in the CEEMD, which represents an adaptive technology of decomposition of the signal into internal functions, called IMF. The peculiarity is that the basic functions used for the decomposition are extracted directly from the source signal, and allow you to take into account only its own peculiarities (hidden modulations, areas of energy concentration, etc.).

As a result of the CEEMD, the finite number of IMFs is extracted from each fragment of the speech signal $x_s(n)$:

$$x_s(n) = \sum_{i=1}^{I-1} IMF_{s,i}(n),$$

where $IMF_{s,i}(n)$ are the received IMFs, $i=1,2,\dots,I$ is a number of an IMF.

Unlike from other methods of decomposition, the peculiarity of the CEEMD is the repeated addition of white noise to the original speech signal with direct and inverse values of amplitude, and the calculation of the average value of IMF as the end true result [5]:

$$\begin{bmatrix} y_{s,j}(n) \\ y_{s,j}(n)^* \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} x_s(n) \\ w_j(n) \end{bmatrix},$$

where $w_j(n)$ is the added white noise, $y_j(n)$ is the sum of the noisy speech signal fragment $x_s(n)$ with the white noise, $y_j(n)^*$ is the sum of the noisy speech signal fragment $x_s(n)$ with the inverse value of the amplitude of the white noise;

$$IMF_{s,i}(n) = \frac{\sum_{j=1}^J IMF_{s,ji}(n)}{J},$$

where $IMF_{s,ji}(n)$ are the IMF received at different decompositions of $y_{s,j}(n)$ and $y_{s,j}(n)^*$ signals, $j=1,2,\dots,J$ is the amount of decomposition cycles (additions to the white noise signal).

A character of amplitude change of the speech signal in time is called the amplitude distribution, which is an important informative speech parameter. Changes of the signal amplitude are described with the help of a short-term signal energy function. Energy calculation of the IMF fragment is done by the formula:

$$E_{s,i}(n) = \sum_{n=1}^N [IMF_{s,i}(n)]^2,$$

where $E_{s,i}$ is the IMF energy of the speech signal fragment.

The human auditory apparatus captures speech nonlinearly; the differences between the energy of sites of the useful signal and the pause must be significant, for the human to capture the amplitude change. To compress the amplitude of the signal with a large dynamic range taking the logarithm of energy of fragments is applied maximally closing the algorithm operation to the functional of the human auditory apparatus [10].

According to the physiological aspect of speech formation, before the pronunciation of the voice command a human makes the momentary pause of typically 200 ms or more. This site does not contain speech and corresponds to the silence with the background noise. Using the averaged values of the IMF energy logarithms, the threshold values of the energy logarithms for the detection of the boundary of the useful signal and the pause can be generated. Defining the threshold values of IMF energy logarithms is carried out according to the formula:

$$LE_{i,thres.} = \frac{1}{10} \sum_{s=0}^9 LE_{IMFs,i},$$

where $LE_{i,thres.}$ is a threshold value of the IMF energy logarithms, $LE_{IMFs,i}$ is the IMF energy logarithm.

Then, the values of the IMF energy logarithms of the remaining fragments are compared with the threshold values $LE_{s,IMFi} \geq LE_{IMFi,thres.}$. If the condition is satisfied, then the fragment is considered to be a signal $s=S_{speech}$, and if the condition is not satisfied, then the fragment is considered to be a pause $s=S_{pause}$.

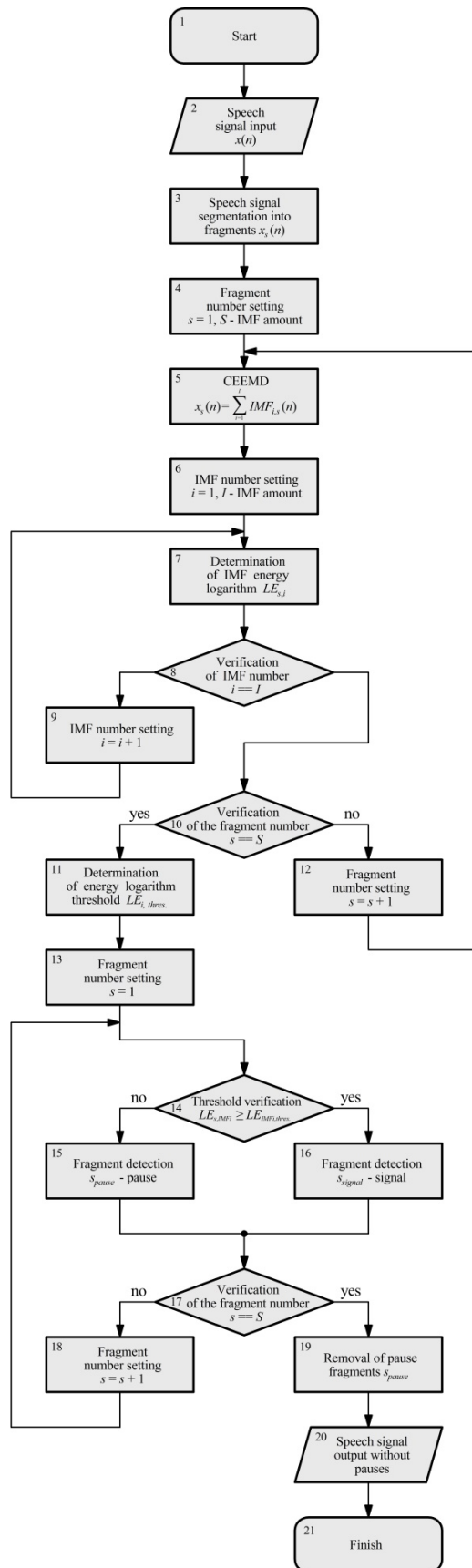


Fig. 1. Block diagram for speech/pause detection algorithm

III. STUDY OF THE DEVELOPED SPEECH/PAUSE DETECTION ALGORITHM

To study the algorithm a test sample of 500 pure speech signals, comprising monosyllabic and polysyllabic phrases, has been generated. Each test signal has been added the additive background white noise. Changing the level of noise for each test signal, speech signals with signal-to-noise ratio values (SNR) from 10 to 35 dB and an increment of 5 dB have been received.

The actual detection rate has been used as a criterion of efficiency of the developed segmentation algorithm. Detection rate (DR) is dimensionless quantity, equal to the ratio of correct detection of the signal fragments to the total number of fragments ($DR_{speech}, \%$).

The results of the study were evaluated in comparison with the known speech/pause detection algorithms (Table I).

TABLE COMPARATIVE ANALYSIS OF THE DEVELOPED AND WELL-KNOWN ALGORITHMS.

SNR _{IN} , dB	DR _{speech} , %			
	STE+ZCR	IE	MFCC	Developed algorithm
10	67.1	73.2	76.2	71.3
15	74.9	80.5	81.4	84.3
20	79.3	82.1	84.5	87.6
25	82.6	87.4	88.1	92.5
30	88.3	90.3	90.7	94.3
35	90.2	92.4	92.1	95.2

As seen from the results, the developed algorithm provides the best result especially with low SNR values. Thus, the use of the developed algorithm in VCS will allow significantly increasing the efficiency of speech/pause detection and the recognition accuracy of speech commands.

IV. CONCLUSION

A speech/pause detection algorithm has been developed. A block diagram for the algorithm with a detailed mathematical description is presented. The results of studies compared with the known algorithms based on STE+ZCR, IE and MFCC are given. In accordance with the results the developed algorithm enhances the actual detection rate by the average of 6%. A comparative analysis of the research results shows that the algorithm may find practical application in VCS used in a noisy environment.

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